Data Flow Diagram (DFD)

SMARTBRIDGE - Data Flow Diagram

Date: 21/09/2026

Team ID: SWTID-2026-4223

Team Size: 4

Team Leader: Sweatha P

Team Members:Lakshmi Narayanan R, Rithika P, Sandhiya K

Project Name: AI FitTrack

Maximum Marks: 2 Marks

DFD Symbol Legend

External Entity: An entity outside the system that sends or receives data (e.g., Client Application, Google Gemini API).

Process: An activity that transforms data (e.g., 1.0 User Auth, 2.0 Workout Manager, 3.0 Gemini AI Service).

Data Store: A repository holding data for later use (e.g., D1 Users Collection, D2 Workouts Collection).

Data Flow: Represents the movement of data between entities, processes, and stores.

DFD Data Flow Specifications

API Client / User → 1.0 Auth Engine: Reg/Login Request

API Client / User → 2.0 Workout Manager: Log Workout

API Client / User → 3.0 AI Service Manager: AI Guidance Request

1.0 Auth Engine → Auth Middleware: Validate

1.0 Auth Engine → D1 Users Store: Write User

2.0 Workout Manager → D2 Workouts Store: Query/Store

3.0 AI Service Manager → D2 Workouts Store: Fetch Stats

3.0 AI Service Manager → Google Gemini API: Send Prompt

Google Gemini API → 3.0 AI Service Manager: AI Response

3.0 AI Service Manager → API Client / User: JSON Recommendation

Process 1.0 (User Authentication): Receives credentials from API Client, hashes password using bcrypt, saves/verifies record in D1: Users Store, and returns JWT bearer token.

Process 2.0 (Workout Management): Authenticates JWT token via middleware, processes CRUD/search requests, and writes/queries records from D2: Workouts Store.

Process 3.0 (Gemini AI Service Integration): Receives user metrics (age, fitness goals, duration, calories), formats prompt, queries Google Gemini API (@google/genai), and returns structured JSON workout plans and fitness insights.